PROJECT PLANNING & MANAGEMENT

Project Proposal

Arabic Sign Language Translator

Objective:

To develop a system that translates Arabic sign language into written Arabic text to help people with hearing and speech impairments communicate more easily with others.

Description:

The project aims to recognize Arabic sign language gestures using computer vision and machine learning techniques. The system will capture hand gestures through a camera, analyze them, and translate the recognized signs into Arabic text. The goal is to make communication more accessible and effective.

Tools & Design:

Jupyter Notebook or Google Colab for model development, a camera for capturing gestures, and a simple user interface for displaying the translated text.

Technologies to Use:

Python, OpenCV, TensorFlow or PyTorch, NumPy, Pandas, and machine learning/deep learning techniques.

Team Members & Responsibilities:

Amira Sameh Mahmoud Mohamed Khalil ElDeeb	Project documentation, testing coordination, and presentation.
Mennatullah Hany Mohamed Hassan	Data preprocessing and preparation.
Naira Hatem Elsherbiny Eid	Model development and evaluation.
Rokaia Ali Bakr El-Gouhari	Dataset collection and organization.
Salma Osama Ahmed Kandil	UI/UX design, Flutter Mobile Application, and system integration

Week 1: Research, Requirements, and Dataset Preparation

- **Rokaia**: Research Arabic Sign Language and define the project objectives and requirements.
- **Naira**: Search for and collect a suitable Arabic Sign Language dataset.
- **Amira**: Explore the dataset, check image quality, labels, and class distribution.
- **Mennatullah**: Research suitable machine learning and deep learning models for sign recognition.
- **Salma**: Review the completed UI/UX design and filters, and plan how the model will integrate with the interface.

Deliverables:

- Defined project objectives and requirements.
- Selected and organized dataset.
- Initial dataset exploration report.
- Proposed machine learning model and technologies.
- Completed UI/UX and filters documented, with an initial integration plan

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Week 2: Data Preprocessing and Initial Model Development

- **Amira:** Clean and organize the dataset and verify the sign labels.
- **Naira:** Apply image preprocessing techniques, such as resizing and normalization.
- **Rokaia:** Split the dataset into training, validation, and testing sets.
- **Mennatullah:** Build and train the initial machine learning or deep learning model.
- **Salma:** Convert UI-UX design into a flutter code.

Deliverables:

- Cleaned and organized dataset.
- Preprocessed images ready for training.
- Training, validation, and testing datasets.
- Initial trained model with preliminary evaluation results.
- UI Flutter Application .

Week 3: Model Evaluation and Application Integration

- **Naira:** Evaluate model performance using accuracy, precision, recall, and a confusion matrix where appropriate.
- **Rokaia:** Analyze prediction errors and help improve the model's performance.
- **Amira:** Test the model on unseen sign language images or video frames.
- **Mennatullah:** Integrate the trained model with the user interface and camera input, if supported by the project scope.
- **Salma:** Test the UI/UX and filters and ensure that the interface displays the predicted Arabic text correctly.

Deliverables:

- Model evaluation results and performance metrics.
- Improved model based on testing results.
- Initial integration of the model with the interface.
- Tested camera or image input functionality, according to the selected implementation.
- Initial working version of the Arabic Sign Language Translator.

Week 4: Testing, Refinement, and Final Delivery

- **Mennatullah:** Perform final tests on the dataset and verify the model's predictions.
- **Naira:** Fix preprocessing issues and document the final data preparation steps.
- **Salma:** Finalize the model and record the final evaluation results.
- **Amira:** Test the complete system, fix integration errors, and improve functionality.
- **Rokaia:** Prepare the final project report, presentation slides, and system demonstration.

Deliverables:

- Final tested version of the Arabic Sign Language Translator.
- Final model evaluation report.
- Integrated and tested user interface and filters.
- Complete project documentation and presentation slides.

Risk Assessment & Mitigation Plan

Risk	Potential Impact	Mitigation
Limited or imbalanced dataset	May reduce recognition performance.	Review class distribution; use suitable augmentation or seek additional data if permitted.
Similar-looking signs or incorrect predictions	May produce incorrect Arabic text.	Analyze misclassified samples and compare suitable model approaches.
Slow inference or integration issues	May make the app unresponsive.	Test deployment early, measure inference time, and optimize if needed.
Lighting and camera variation	May reduce real-world performance.	Test under varied conditions and document limitations.
Schedule delays	May affect prototype completion.	Track tasks weekly, prioritize core features, and reserve time for testing.

Key Performance Indicators (KPIs)

KPI	How It Will Be Measured
Model Accuracy	Percentage of correctly classified test samples.
Weighted F1-score	Balance between precision and recall across classes.
Inference Time	Time required to produce one prediction.
Application Success Rate	Percentage of test attempts completing the recognition flow successfully.
Task Completion Rate	Percentage of planned tasks completed by their deadlines.